# ARJUN TYAGI

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## SUMMARY

- Ph.D. student in Computer Architecture, specializing in cross-stack development and integration of processing-using-memory systems
- Extensive experience in memory systems, from devices to circuits, gained through multiple research internships, with a strong interest in CPU and memory subsystem engineering.

## EDUCATION

**University of Illinois Urbana-Champaign** — **Champaign, IL**
*Ph.D in Computer Science* — *Expected May 2029*
· GPA: 3.88/4.0

**Birla Institute of Technology & Science, Pilani** — **Goa, India**
*Bachelor of Engineering in Electronics and Instrumentation Engineering* — *Aug. 2019 - May 2023*
· GPA: 8.99/10.0

## PUBLICATIONS, PREPRINTS & TUTORIALS

**Arjun Tyagi**, Minh S. Q. Trung, Ankit Shukla, Shaloo Rakheja, and Saugata Ghose, *[Title Redacted]*, To be submitted to 2026 ACM/IEEE Design Automation Conference (DAC)

**Arjun Tyagi** and Shubham Sahay, *Efficient Reinforcement Learning On Passive RRAM Crossbar Array*, Submitted to IEEE Transactions on Very Large Scale Integration Systems (TVLSI), preprint available upon request

**Arjun Tyagi** and Shahar Kvatinsky, *Assessing the Performance of Stateful Logic in 1-Selector-1-RRAM Crossbar Arrays*, 2024 IEEE International Symposium on Circuits and Systems (ISCAS), pp. 1-5, doi: 10.1109/ISCAS58744.2024.10558539

Ben Feinberg, T. Patrick Xiao, Minh S. Q. Truong, **Arjun Tyagi**, Ryan Wong, Yiqiu Sun, and Saugata Ghose, *Tutorial on Simulation for Processing-Using-Memory Systems*, 2024 IEEE International Symposium on Computer Architecture (ISCA)

## RESEARCH EXPERIENCE

ARCANA Research Group, PI: Prof. Saugata Ghose — **Urbana, IL**

***Rapid, Accurate Chip-Scale Simulation of Digital Compute Using Emerging Memory Devices*** — *Sept. 2023 - present*

- Developed a fast simulation framework for digital in-memory computing, enabling chip-scale modeling with $> 10,000\times$ speedup over SPICE tools while maintaining high accuracy
- Introduced techniques such as memoization and dynamic time stepping to drastically reduce simulation time and memory usage while still being robust to device variability
- Enabled seamless integration of Verilog-A device models, supporting iterative co-design across device, circuit, and architecture levels
- Incorporated power and thermal modeling, along with mechanisms to capture reliability degradation in emerging memory and logic devices

ASIC$^2$ Research Group, PI: Prof. Shahar Kvatinsky — **Haifa, Israel**

***Assessing the Performance of Stateful Logic in 1-Selector-1-RRAM Crossbar Arrays*** — *Jan. - Jun. 2023*

- Developed a compact Verilog-A model for 1S1R memory cells, aimed for large-scale simulations

- Modeled parasitic RC effects for intra-array interconnects to realistically evaluate the effect of array size on performance degradation
- Simulated passive (1R) and active (1S1R) RRAM arrays of varying sizes ($4 \times 4$ to $512 \times 512$) for in-memory computing using MAGIC logic gates
- Demonstrated that 1S1R arrays offer up to 24.4× lower power and 4.5× higher readout margin than 1R arrays, enabling reliable scaling to large arrays

***CtrlPIM: A Microcode-Based Controller for Memristive Memory Processing Unit*** *Jan. - Jun. 2023*

- Co-designed a memory controller to support a PIM ISA for memristive memory processing unit (mMPU)
- Implemented the controller on an FPGA consisting of Zynq-7000 SoC

NeuroChaSe Group, PI: Prof. Shubham Sahay **Kanpur, India**

***Efficient Reinforcement Learning on Passive RRAM Crossbar Array*** *Jan. - Jul. 2022*

- Performed an algorithm-hardware co-design and proposed a novel implementation of Monte Carlo reinforcement learning algorithm on passive RRAM crossbar array
- Demonstrated $> 100,000\times$ area reduction over prior digital and 1T1R-based implementations, while maintaining robustness to spatial/temporal variations and endurance failures in RRAM devices

## AWARDS & SCHOLARSHIPS

- **First-Year Ph.D. Fellowship** *Aug. 2024 - May 2025*
  University of Illinois Urbana-Champaign
- **Summer Undergraduate Reseach Internship** *May - Aug. 2022*
  Arizona State University
- **Summer Research Fellowship Programme** *May - Aug. 2022*
  Indian Institute of Technology Delhi

## SKILLS & ABILITIES

- **Languages/Applications**: C, C++, Python, OpenMP, Verilog HDL, Verilog A, MATLAB
- **Graduate Courses**: Computer System Organization, Parallel Computer Architectures, Emerging Memory & Storage Systems, Parallel Programming, Programming Languages, Compiler Construction
- **Undergraduate Courses**: Computer Architecture, Microprocessor & Interfacing, Microelectronic Circuits, Analog & Digital VLSI Design, Analog Electronics, Digital Design

## TUTORING & VOLUNTEERING EXPERIENCE

- **Graduate Student Ambassador at UIUC CS** *Mar. 2025*
- **Undergraduate/Graduate Mentor**, UIUC
  · Dylan Meeks, BS ECE '26 at University of Texas at Austin *Jun. - Aug. 2025*
  · Qingyuan Liu, MS ECE '26 at University of Illinois Urbana-Champaign *Aug. 2025 - present*

## TEACHING EXPERIENCE

- **ECE 313: Analog & Digital VLSI Design** **BITS Pilani**
  *Undergraduate Teaching Assistant* *Aug. - Dec 2022*
- **ECE 241: Microprocessor & Interfacing** **BITS Pilani**
  *Undergraduate Teaching Assistant* *Jan. - May 2022*
- **ECE 216: Digital Design** **BITS Pilani**
  *Undergraduate Teaching Assistant* *Aug. - Dec 2021*